

$$I_c: [A]_0, [B]_0, [C]_0$$

$$\begin{bmatrix} \dot{A} \\ \dot{B} \\ \dot{C} \end{bmatrix} = \begin{bmatrix} -k_1 & k_3 & 0 \\ k_1 & -k_3 & -k_2 \\ 0 & 0 & k_2 \end{bmatrix} \begin{bmatrix} A \\ [B][C] \\ [B]^2 \end{bmatrix}$$

~~$$\begin{bmatrix} \dot{A} \\ \dot{B} \\ \dot{C} \end{bmatrix} = \begin{bmatrix} -k_1 & k_3 & 0 \\ k_1 & -k_3 & -k_2 \\ 0 & 0 & k_2 \end{bmatrix} \begin{bmatrix} A \\ [B][C] \\ [B]^2 \end{bmatrix}$$~~

By QSS:

$$\begin{bmatrix} \dot{A} \\ \dot{C} \end{bmatrix} = \begin{bmatrix} -k_1 & k_3 & 0 \\ 0 & 0 & k_2 \end{bmatrix} \begin{bmatrix} [A] \\ [B][C] \\ [B]^2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{A} \\ \dot{C} \end{bmatrix} = \begin{bmatrix} -k_1 & k_3[B] \\ 0 & 0 \end{bmatrix} \begin{bmatrix} [A] \\ [C] \end{bmatrix} + \begin{bmatrix} 0 \\ k_2[B]^2 \end{bmatrix}$$

$$\dot{X} = \underline{A} X + b$$

eigen-decompose matrix $\underline{A}$: $(-k_1 - \lambda) \cdot 0 - k_3[B] \cdot 0 = 0$ (singular matrix)

$$\dot{A} = -k_1[A] + (k_3[B]) \cdot [C] \quad (1)$$

$$\dot{C} = k_2[B]^2 \rightarrow C = k_2[B]^2 \cdot t + C_0 \quad (2)$$

$$\begin{aligned} (1): \quad \frac{d[A]}{dt} &= -k_1[A] + (k_3[B]) \left(k_2[B]^2 \cdot t + C_0 \right) \\ &= \underbrace{-k_1[A]}_{\alpha} + \underbrace{k_2 k_3 [B]^3}_{\beta} t + \underbrace{k_3 [B] C_0}_{\gamma} \end{aligned}$$

$$\frac{d[A]}{dt} = \alpha[A] + \beta t + \gamma$$

$$\frac{d[A]}{dt} - \alpha[A] = \beta t + \gamma$$

$$e^{-\alpha t} \frac{d[A]}{dt} - \alpha e^{-\alpha t} [A] = \beta t e^{-\alpha t} + \gamma e^{-\alpha t}$$

$$\int_0^t \frac{d}{dt} (e^{-\alpha t} [A]) dt = \beta \int_0^t t e^{-\alpha t} dt + \gamma \int_0^t e^{-\alpha t} dt$$

~~$$e^{-\alpha t} [A]_t - [A]_0 = \beta \int_0^t t e^{-\alpha t} dt + \gamma \int_0^t e^{-\alpha t} dt$$~~

$$e^{-\alpha t} [A]_t - [A]_0 = -\frac{\beta}{\alpha} t e^{-\alpha t} + \left(\frac{\beta}{\alpha} + \gamma \right) \int_0^t e^{-\alpha t} dt$$

$$\begin{aligned}
 e^{-\alpha t} [A]_t - [A]_0 &= -\frac{\beta}{\alpha} t e^{-\alpha t} + \left(\frac{\beta}{\alpha^2} + \gamma\right) (-\frac{1}{\alpha})(e^{-\alpha t} - 1) \\
 &= -\frac{\beta}{\alpha} t e^{-\alpha t} - \left(\frac{\beta}{\alpha^2} + \gamma\right) (e^{-\alpha t} - 1) \\
 &= -\frac{\beta}{\alpha} t e^{-\alpha t} - \left(\frac{\beta}{\alpha^2} + \gamma\right) e^{-\alpha t} + \left(\frac{\beta}{\alpha^2} + \gamma\right)
 \end{aligned}$$

$$e^{-\alpha t} [A]_t = -\frac{\beta}{\alpha} t e^{-\alpha t} - \left(\frac{\beta}{\alpha^2} + \gamma\right) e^{-\alpha t} + \left(\frac{\beta}{\alpha^2} + \gamma + [A]_0\right)$$

	$[A]_t = -\frac{\beta}{\alpha} t - \left(\frac{\beta}{\alpha^2} + \gamma\right) + \left(\frac{\beta}{\alpha^2} + \gamma + [A]_0\right) \cdot e^{\alpha t}$
together with:	$[B]_t = [B]$ $[C]_t = [B]^2 \cdot t + [C]_0$

$$\begin{aligned}
 [B]_0 &= 0 : \quad \beta = 0, \quad \gamma = 0, \\
 [A]_t &= [A]_0 \cdot e^{\alpha t} \\
 [B]_t &= 0 \\
 [C]_t &= 0
 \end{aligned}$$

$$\frac{\beta}{\alpha} = -\frac{k_2 k_3 [B]^3}{k_1}, \quad \frac{\beta}{\alpha^2} + \gamma = \frac{k_2 k_3 [B]^3}{k_1^2 [A]^2} + \frac{k_3 [B] [C]_0}{-k_1},$$

$$\frac{\beta}{\alpha^2} + \gamma + [A]_0 = \frac{k_2 k_3 [B]^3}{k_1^2} + \frac{k_3 [B] \cdot C_0}{-k_1} + 1,$$

~~Q38A2~~ ~~at~~ ~~blows away~~